ASSIGNMENT-2

EX 1.

Evaluate the following expressions:

(i) $\int_{-\infty}^{\infty} cos\frac{\pi}{2}(t-5))\delta(2t-3)dt$

(ii) $e^{2t}\cos\left(\frac{50}{\pi}t\right)\delta(t+1)$

(iii) $\int_{-\infty}^{\infty} e^{2t}\cos\frac{\pi}{2}\left(\frac{50}{\pi}t\right)\delta(t+1)dt$

EX 2.

A continuous-time signal $x(t)$ is shown in Fig. 1-27. Sketch and label each of the following signals.

(a) $x(t)u(1-t)$; (b) $x(t)[u(t)-u(t-1)]$; (c) $x(t)\delta(t-\frac{3}{2})$

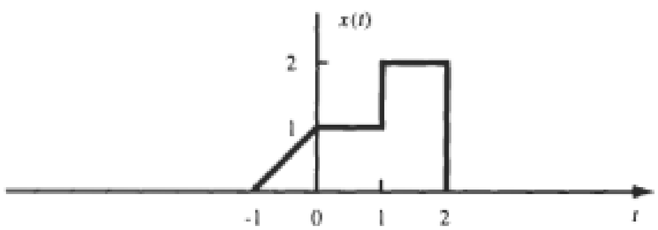

Fig. 1-27

Ex 3.

A discrete-time signal $x[n]$ is shown in Fig. 1-29. Sketch and label each of the following signals.

(a) $x[n]u[1-n]$; (b) $x[n]\{u[n+2]-u[n]\}$; (c) $x[n]\delta[n-1]$

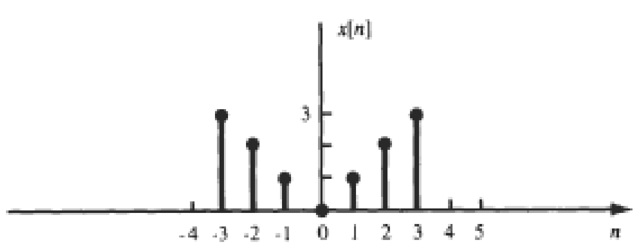

Fig. 1-29

EX 4.

A system has the input-output relation given by

$$y[n] = \mathbf{T}\{x[n]\} = nx[n] \tag{1.113}$$

Determine whether the system is (*a*) memoryless, (*b*) causal, (*c*) linear, (*d*) time-invariant, or (*e*) stable.

Ex5.

Consider a continuous-time system with the input-output relation

$$y(t) = \mathbf{T}\{x(t)\} = \frac{1}{T}\int_{t-T/2}^{t+T/2} x(\tau)\, d\tau$$

Determine whether this system is (*a*) linear, (*b*) time-invariant, (*c*) causal.